



AgriGaurd

**Smart Irrigation and Water Management
System**

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Problem Statement...

1. Background and Problem Statement

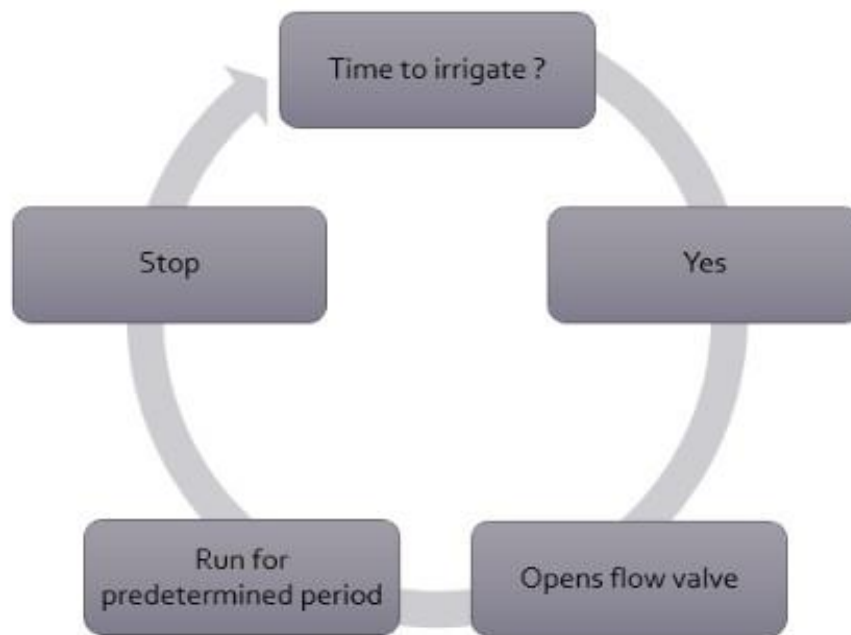
Agriculture is fundamentally dependent on water. However, water availability is becoming increasingly uncertain due to population growth, climate variability, drought and increasing competition for freshwater resources.

Agriculture currently accounts for approximately **72%** of global freshwater withdrawals, with irrigation being the primary driver of agricultural water demand. At the same time, approximately **1.2 billion people** live in agricultural areas experiencing severe water constraints. The Food and Agriculture Organization (FAO) estimates that by 2050 agriculture will need to produce approximately **50%** more food, feed and fibre than in 2012, while freshwater demand is expected to increase by another **25%**.

This creates a fundamental challenge that agriculture needs to produce more with less water. The problem is therefore not simply a shortage of water. It is also a problem of *“how available water is managed and allocated.”*

The Irrigation Problem

Traditional irrigation decisions are often based on fixed schedules, timers, manual observation or farmer experience. A simplified conventional approach would look like:



The problem with this approach is that the environment is not fixed, soil moisture changes continuously, temperature fluctuates, humidity changes, rainfall changes, different soils retain water differently, different zones/areas dry at different rates.

Consequently, applying the same irrigation schedule to every zone can result in:

- Under-irrigation
- Over-irrigation
- Unnecessary water consumption
- Nutrient leaching
- Waterlogging
- Reduced crop performance
- Increased operating costs

The central problem identified by AgriGuard is that farmers need to make irrigation decisions using changing field conditions, but many low- and mid-scale operations lack affordable systems capable of continuously measuring, interpreting and acting on those conditions.

How Much Water Is Being Lost

Irrigation losses occur for many reasons, including evaporation, runoff, deep percolation, poor infrastructure and inappropriate irrigation scheduling.

FAO (***Farm and Agriculture Organization***) indicates that **40%** or more of water diverted for irrigation can be lost at farm level in some circumstances, through mechanisms such as deep percolation and surface runoff.

This does not mean that 40% of all irrigation water is universally wasted by farmers. Rather, it demonstrates the magnitude of potential losses throughout irrigation systems.

Research provides stronger evidence that better scheduling can directly reduce water use. A systematic review of soil-water-sensor irrigation scheduling research found that sensor-based scheduling used, on average:

Comparison	Average water reduction
Traditional scheduling	38%
Computer-model scheduling	16%
Evapotranspiration replacement	20%
Canopy-temperature methods	1%
Traditional scheduling	38%

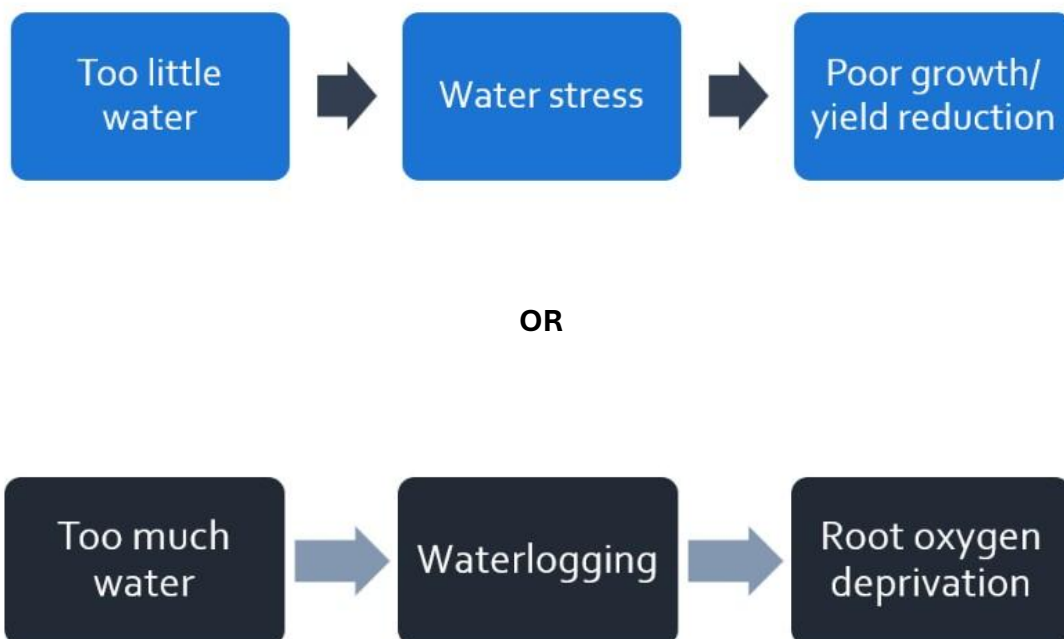
The review also found that soil-water-sensor scheduling produced **24%** higher yields on average compared with traditional scheduling in the relevant comparisons. This provides an important foundation for AgriGuard, *better information can result in better irrigation decisions.*

Overwatering Is Also a Problem

Water conservation cannot simply mean reducing irrigation as much as possible. Plants require water, but they also require oxygen around their roots. When soil remains saturated for extended periods, oxygen availability to the root system decreases. This can impair root growth, nutrient uptake and plant development.

A global meta-analysis containing **2,419 comparisons from 115 studies** found that waterlogging reduced crop yield by an average of approximately **32.9%**.

Therefore, both extremes are problematic:



This leads to one of the core principles behind AgriGuard:

*The objective is not to give crops **more water**, but it is to give crops the **right amount** of water at the right time.*

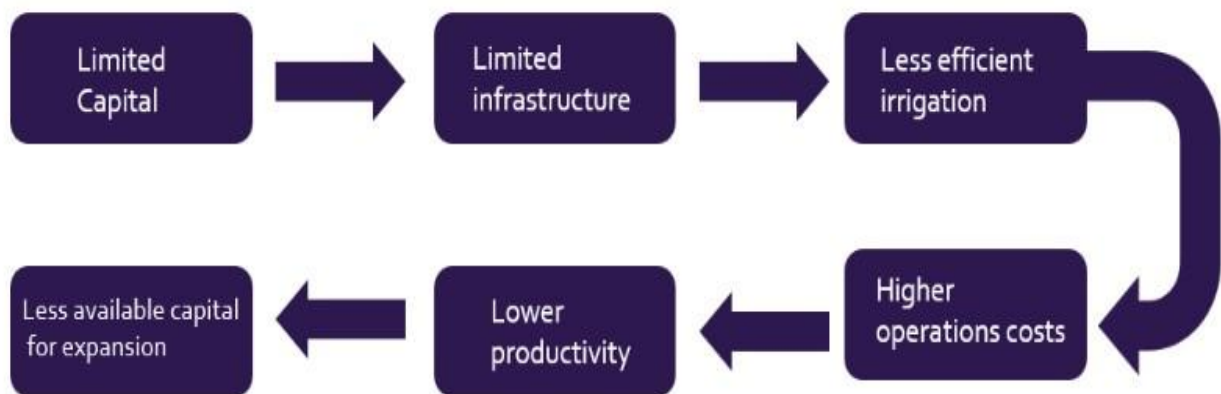
Why Small-Scale Farmers Are Particularly Affected

The irrigation problem becomes more difficult when considering the economic and environmental conditions faced by small and mid-scale farmers.

A 2025 review specifically examining smallholder irrigation development in South Africa identified several constraints, including:

- Limited access to irrigation water
- Inadequate infrastructure
- Drought
- Limited land access
- Financial constraints
- Technological limitations
- Knowledge and skills gaps
- Challenges implementing water policies
- Climate-change impacts

These constraints interact with one another.



This creates a cycle that makes it difficult for small-scale farmers to transition toward more advanced agricultural operations.

Economic Constraints

Small-scale farmers commonly face barriers associated with:

- **Initial capital:**

Advanced irrigation systems can require pumps, pipes, valves, sensors, controllers and installation.

- **Operating costs:**

Energy, water, maintenance and replacement components add recurring costs.

- **Access to finance:**

Agricultural lending can be difficult because small farms may lack collateral, have small transaction sizes and face high production and market risks.

Research on agricultural finance in Africa notes that less than **4% of total commercial bank lending** goes to agriculture, with factors such as lack of collateral, remoteness, infrastructure limitations and variable rainfall contributing to lending difficulties.

- **Technology cost:**

Advanced precision-agriculture systems can provide excellent functionality, but their cost and complexity can make them difficult to adopt at smaller scales.

Environmental Constraints

Small-scale farmers are also exposed to environmental uncertainty.

These include:

- Drought
- Unpredictable rainfall
- Increasing temperatures
- Evaporation
- Soil variability
- Water scarcity
- Changing crop-water requirements

Climate change makes the problem more difficult because water supply becomes less predictable while crop water demand can increase. The FAO identifies climate change as a factor reducing the reliability of water supply while increasing crop water requirements. Therefore, a farmer cannot reliably depend on a fixed irrigation schedule throughout an entire growing season.

Research Supporting the AgriGuard Concept

AgriGuard's architecture was influenced by several areas of existing research.

Research strongly supports the use of soil-water sensors for irrigation scheduling.

A 2023 systematic review analysed **97 papers**, including **48 studies** involving soil-water-sensor irrigation scheduling. The review found that sensor-based scheduling reduced water use while maintaining or improving yield in many comparisons.

Importantly, the review also identified *calibration as one of the major factors influencing sensor accuracy*. This directly influenced AgriGuard's decision not to treat a raw sensor ADC data as an automatically valid percentage.

“A percentage is only useful if we know what 0% and 100% actually mean.”

Study on Smart Irrigation

Study into smart irrigation has expanded significantly.

A 2026 systematic review of IoT-enabled smart irrigation systems examined 50 studies and reported water savings ranging from approximately **9–50%** across reviewed systems relative to conventional irrigation. The same review identified high cost, scalability and farmer training among major adoption barriers.

This is significant because it demonstrates both sides of the problem, which is that the technology works, but also the technology must also be accessible.

AgriGuard therefore attempts to combine the advantages of smart irrigation with a low-cost embedded architecture.

Introducing AgriGuard

AgriGuard is an **ESP32-based smart irrigation and water-management system** designed for low to mid-scale agricultural operations.

The system consists of:

- ✓ ESP32 MCU
- ✓ 3 × capacitive soil-moisture sensors
- ✓ 3 × solenoid irrigation valves
- ✓ 4-channel relay module
- ✓ Water-level sensor
- ✓ LCD I2C
- ✓ ON/OFF switch
- ✓ Emergency-stop switch
- ✓ Reset switch
- ✓ Pull-up resistors

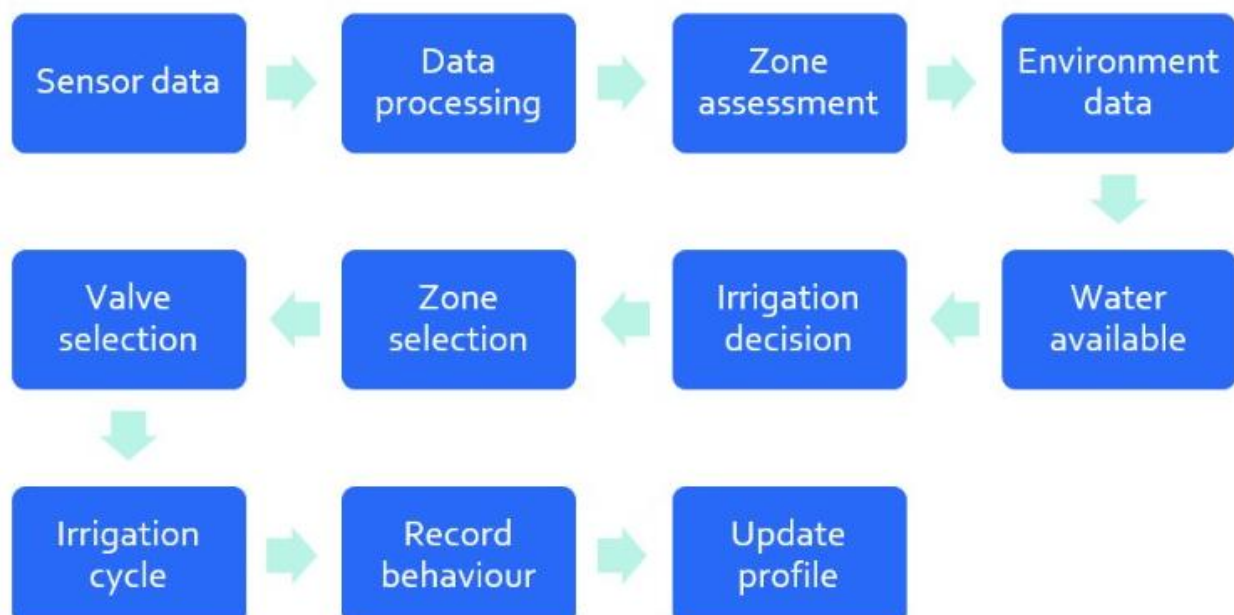
The prototype is organised around three irrigation zones.

Zone 1 → **Soil Sensor 1** → **Valve 1**

Zone 2 → **Soil Sensor 2** → **Valve 2**

Zone 3 → **Soil Sensor 3** → **Valve 3**

The ESP32 acts as the central processing and control unit. How AgriGaurd works:



. What AgriGuard Considers

AgriGuard is designed to consider several factors simultaneously.

- ✓ ***Current soil moisture:***

Determines the current soil moisture of each zone.

- ✓ ***Soil type***

Considers the differences in how different soils retain and release water.

- ✓ ***Drying rate***

Determines how the rate at which each zone is losing moisture.

- ✓ ***Temperature***

Considers how cold or hot the area is can influence the drying rate of the farm.

- ✓ ***Humidity***

Looks at the moisture within the air to determine the amount of water that has left.

- ✓ ***Rainfall/weather***

Can influence whether irrigation should be delayed or adjusted.

- ✓ ***Available water***

Determines whether there is sufficient water in the tank for irrigation.

- ✓ ***Historical irrigation cycles***

Provides information about how the zone responded to previous irrigation events.

- ✓ ***Zone priority***

Allows available water to be allocated according to the current needs of different zones

AgriGuard and Small-Scale Farmer Constraints

AgriGuard is designed specifically to help resolve the issues farmers were encountering.

Farmer constraint	AgriGuard response
Limited capital	Low-cost ESP32-based architecture
Limited irrigation infrastructure	Modular zone-based design
Water scarcity	Water-level monitoring and water budgeting
Unpredictable weather	Weather-assisted decision-making
Different soil types	Zone-specific calibration
Different drying rates	Drying-rate analysis
Limited technical resources	Local LCD interface
Safety concerns	Physical E-STOP and reset
Internet unreliability	Local sensor operation remains available
Need for expansion	Modular zone architecture
Limited historical data	System progressively builds irrigation profiles

AgriGuard is not attempting to solve every agricultural problem, instead, it focuses on a specific and controllable problem. Improving irrigation decision-making and water allocation.

Innovation

The biggest innovation in AgriGuard is not simply automatic irrigation, it is the system's development in *learning-based water budget*.

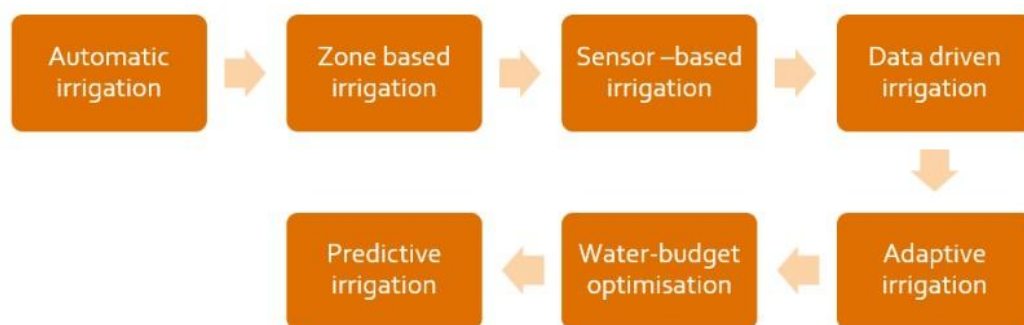
The system can progressively learn:

- How quickly each zone dries
- How much irrigation each zone normally requires
- How long irrigation cycles run
- How environmental conditions influence drying
- How much water is available
- How much water different zones require

This creates the possibility of moving from, “Turn the valve on when the soil is dry”, to determining on what I have learned about this zone, the current environmental conditions and the amount of water available, this zone should receive priority.

The long-term vision is to create a ***self-improving irrigation system***.

The conceptual progression is:



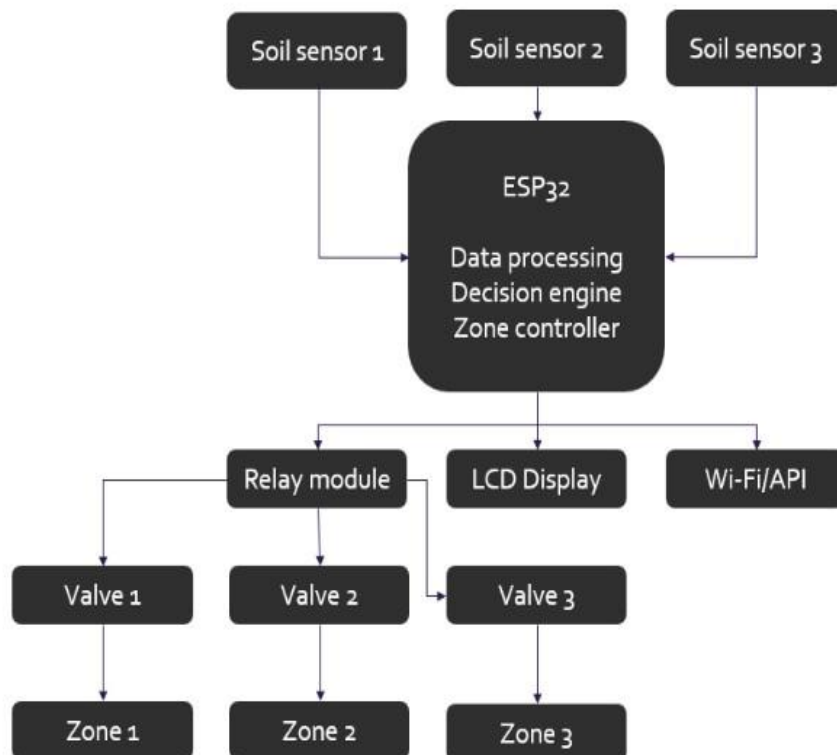
Future versions could use machine-learning models to predict soil moisture using historical:

- Soil moisture
- Temperature
- Humidity

- Weather
- Irrigation duration
- Water usage

Recent research is already investigating exactly this direction: soil-moisture forecasting models using historical sensor, weather and irrigation observations. The major change would be that AgriGuard would no longer wait for soil to become sufficiently dry before reacting, it would predict when a zone is likely to require water and prepare the water budget before the demand occurs.

System Architecture



Water-Level Sensor —► ESP32

ON/OFF —► ESP32

E-STOP —► Safety Logic

RESET —► ESP32

Soil-Monitoring Subsystem

Three capacitive sensors provide the soil-moisture information. Each sensor is assigned to one zone. The raw analogue signal is processed by the ESP32 and converted using calibration information. The system can then monitor changes over time. This provides the basis for calculating drying rates.

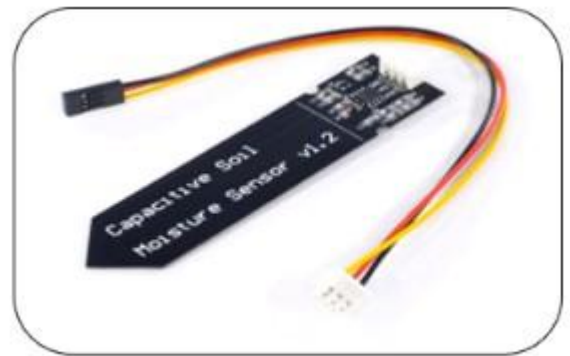


Figure 1 Capacitive soil moisture sensor

Irrigation-Control Subsystem

The ESP32 is the brains of the entire operations, it handles all logical, arithmetic calculation and all processing the system will perform. The ESP32 is a (Come up with something explain how the ESP32 is used in the system).



Figure 2 ESP32 Dev board

Relay module

The ESP32 cannot directly drive the irrigation valves. The relay module therefore acts as the interface between the controller and the valve circuits. (Say something about how the relay works and how it is used in this system).



Figure 3 4-Channel relay module

Battery

Provides power to the entire system.

Figure 4 12V Battery



Water-Level Subsystem

The ultrasonic water-level sensor provides information about the available water supply.

This becomes an input into the water-budget system. (Same something about how the ultrasonic water level sensor is used in the system)



Figure 5 Ultrasonic water-level sensor

Weather Subsystem

The ESP32's Wi-Fi capability allows the system to retrieve external weather information.

Potential inputs include:

- Temperature
- Relative humidity
- Precipitation
- Precipitation probability

Weather information can modify the irrigation decision but does not replace local soil measurements. (Talk about how we use open-Mateo to get weather data and how using it is way better for our system)



Figure 6 Open-Mateo weather API

Flo sensor Subsystem

With a flow sensor valve, we can determine the amount of water that was used in a specific zone. With that information we can predict *how much water will zone 1 need in the next hour*. Flow sensor also provides the ability to control irrigation of each zone.



Figure 7 Water flow sensor

User Interface Subsystem

The LCD provides local feedback.

The operator can view information such as:

ZONE 1: 43%

ZONE 2: 67%

ZONE 3: 51%

TANK: 74%

STATUS: NORMAL

The display allows the farmer to interact with the system without requiring a computer.

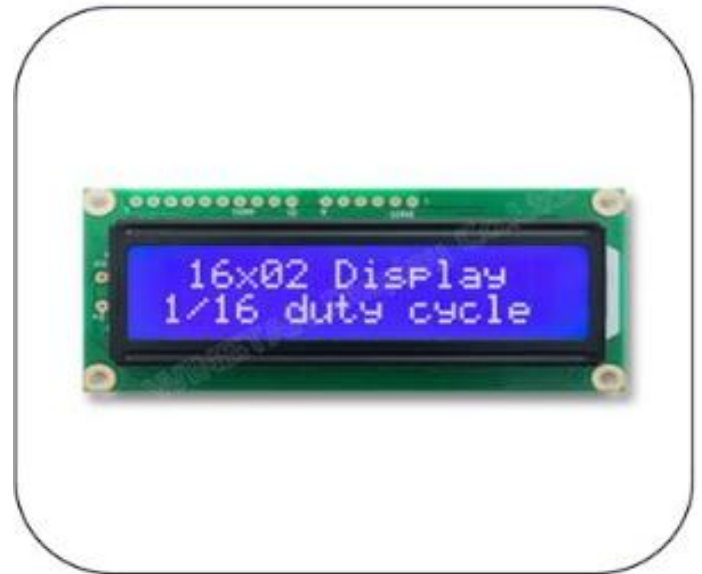


Figure 8 LCD display board

Safety Subsystem

AgriGuard incorporates three physical controls

ON/OFF

Controls normal operation.

E-STOP

Places the system into a safe state.

RESET

Allows controlled recovery.

The intended sequence is:

Normal Operation → Emergency Detected → E-STOP → Safe State → Problem-Corrected → RESET → Normal Operation



Figure 9 ON/OFF latch and E-STOP switches

Development Challenges

Building AgriGuard was not simply a matter of connecting sensors to an ESP32 then everything is working. Several problems were encountered during development of the system.

Sensor Calibration

Problem

Raw capacitive sensor values do not automatically translate into meaningful moisture percentages that we can use. Different soil types produce different readings.

Solution

A calibration approach was developed using dry and wet reference conditions. The user first places the sensors in fully dry soil to measure the soil moisture. We use that value as our reference point, then the user again places the sensor in fully wet soil to get our max moisture reading which we will use as our reference point. After we have captured the results, we can now properly convert the raw values from sensor into usable percentages we use to represent different soil moisture levels. In this system, 0% represents fully dry soil and 100% represents fully wet soil. With this knowledge, understanding the sensor readings has become easier. This was one of the reasons the project moved away from treating a raw sensor reading as an absolute percentage.

Different Soil Types

Problem

Sandy, loamy and clay soils behave differently.

The challenge with dealing with different soil types is that each one has their own specific needs and requirements to thrive. Now knowing that, we cannot simply have a fixed threshold we use to irrigate the soil. The main factors we had to consider was the rate at which water was being retained, how fast does water leave the soil and how does temperature affect the soil.

Solution

Classify areas in the farm as zones that have different soil types. With each zone, we provide to its specific needs rather than just having a *one size fits all solution*. To

accommodate each zones specific needs, we looked at each zones characteristics but mainly we focused on:

- ✓ Soil type
- ✓ Dry calibration
- ✓ Wet calibration
- ✓ Moisture history
- ✓ Drying behaviour

Result

The system can progressively treat each zone according to its own behaviour, and also because our system has the capability of knowing what each raw value from sensor means in terms of moisture readings, we placed soil moisture sensors in each zone so we can know what each zone needs. With this technology being able to know each soil moisture of each **zone** has elevated the dynamic of AgriGaurd.

Temperature Effects

Problem

Temperature affects water loss and therefore changes how quickly soil can dry. Using soil moisture alone does not explain why the same zone may dry faster on one day than another.

Solution

Temperature was introduced as an additional environmental factor. To properly determine when to irrigate, we had to refactor our irrigation logic by incorporating weather data into our irrigation logic. We used a platform called Open-Mateo to get all weather data needed to improve how we know when to irrigate.

Threshold Determination

Problem

Since we knew that having a fixed moisture threshold is easy to implement but does not consider for:

- 1) Soil type
- 2) Weather
- 3) Temperature
- 4) Drying rate
- 5) Historical irrigation
- 6) Water availability

Solution

The system evolved toward dynamic decision-making.

Instead of having:

Moisture < 40%

Irrigate

The system now considers:

- ✓ Moisture
- ✓ Drying Rate
- ✓ Environment
- ✓ Weather
- ✓ Water Availability
- ✓ Zone Profile
- ✓ Decision

Water-Budget Development

Problem

Initially, the system could determine whether a zone needed water, but this did not consider “*what happens when there isn't enough water for every zone?*”

Solution

The water-budget concept was introduced.

The system can monitor the available tank water and compare it with estimated zone requirements. How it works is that it looks at all zones and it determines in which state each zone is in. For instance, a zone could either be:

- ❖ Healthy
- ❖ Moderate.
- ❖ Critical.

After determining the state of each zone, it rations the water based on the state its in, if a zone is critical, it will receive majority of the water, the moderate one will receive what's left and the healthy one won't be irrigated. This system gives water priority to zones that in need of water than others. This transformed the project from a simple irrigation controller into a **water-management system**.

Safety Integration

Problem

Software-controlled irrigation creates a safety concern. If the software behaves incorrectly, the operator needs a physical method of stopping the system.

Solution

A dedicated latching E-STOP and reset mechanism was incorporated into the architecture.

The E-STOP is given priority over normal irrigation commands. This ensures that:

The safety system does not depend entirely on the irrigation algorithm functioning correctly.

Hardware Packaging

Problem

Agricultural electronics cannot simply be left exposed.

The system must be protected against:

- Moisture
- Dust

- Soil
- Physical damage
- Outdoor environmental conditions

Solution

A dedicated enclosure is being designed using CAD and can be fabricated using 3D printing.

The enclosure provides:

- Component protection
- Cable management
- Mounting points
- Access to controls
- LCD visibility
- Serviceability

This also allows the physical design to be modified as the prototype evolves.

Overall Impact

AgriGuard addresses the irrigation problem from several directions simultaneously.

Environmental impact

- Reduces unnecessary water application
- Encourages more efficient use of available water
- Reduces risk of overwatering
- Supports water-conservation practices

Agricultural impact

- ❖ Provides zone-specific irrigation
- ❖ Maintains better awareness of soil conditions
- ❖ Supports more consistent irrigation decisions
- ❖ Builds historical knowledge of zones

Economic impact

- Uses relatively accessible components
- Reduces dependence on expensive industrial automation
- Can potentially reduce unnecessary water and energy consumption
- Provides a modular architecture that can grow with the farm

Technological impact

- ✓ Combines embedded systems and IoT
- ✓ Uses real-time sensor information
- ✓ Introduces data-driven irrigation
- ✓ Creates a foundation for predictive irrigation

Conclusion

The irrigation problem facing agriculture is not simply a problem of insufficient water. It is also a problem of **how available water is measured, allocated and used**.

Agriculture accounts for approximately **72% of global freshwater withdrawals**, while climate change, increasing demand and water scarcity are placing additional pressure on agricultural systems.

Research demonstrates that poor irrigation management can result in substantial water losses, while overwatering can also negatively affect crop performance. At the same time, studies of soil-water-sensor irrigation demonstrate that better scheduling can substantially reduce water use while maintaining or improving yield.

However, the existence of advanced commercial technology does not automatically solve the problem for every farmer. Small and mid-scale farmers face constraints involving capital, infrastructure, water access, climate, technical knowledge and scalability. South African research specifically identifies these constraints as barriers to smallholder irrigation development.

AgriGuard was developed in response to this gap. The system combines:

soil-moisture monitoring + zone-based irrigation + water-level monitoring + environmental data + irrigation history + safety controls + water budgeting.

Its most important characteristic is its progression beyond basic automation.

A conventional system may ask:

"Is the soil dry?"

AgriGuard aims to ask:

"How dry is this zone, how quickly is it drying, what environmental conditions are affecting it, how much water is available, how has this zone behaved previously, and is irrigation currently the best use of the available water?"

The immediate system provides automated, zone-based irrigation. The water budgeting is what comes next: transforming the system into a platform capable of building irrigation profiles for each zone, learning from previous irrigation cycles, managing a finite water budget and eventually predicting future water requirements.

AgriGuard is therefore not intended to be another device that simply turns irrigation valves on and off. AgriGuard is being developed as an intelligent water-management platform designed to help smaller agricultural operations make better decisions with the water they already have.

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